



Module 13: Concurrency Control

CPE 011 Database Management System

Slides based on “Database Systems: Design, Implementation, and Management 8th Edition” Chapter 10:
Concurrency Control Slides by Rob, P. and Coronel, C.

Concurrency Control with Locking Methods

- Lock
 - Guarantees exclusive use of a data item to a current transaction
 - Required to prevent another transaction from reading inconsistent data
- Lock manager
 - Responsible for assigning and policing the locks used by transactions
- Lock Granularity
 - Indicates level of lock use
 - Locking can take place at following levels: Database, Table, Page, Row, and Field (attribute)

Lock Granularity

- Database-level lock
 - Entire database is locked
- Table-level lock
 - Entire table is locked
- Page-level lock
 - Entire diskpage is locked
- Row-level lock
 - Allows concurrent transactions to access different rows of same table, even if rows are located on same page
- Field-level lock
 - Allows concurrent transactions to access same row, as long as they require use of different fields (attributes) within that row

**FIGURE
10.3**

Database-level locking sequence

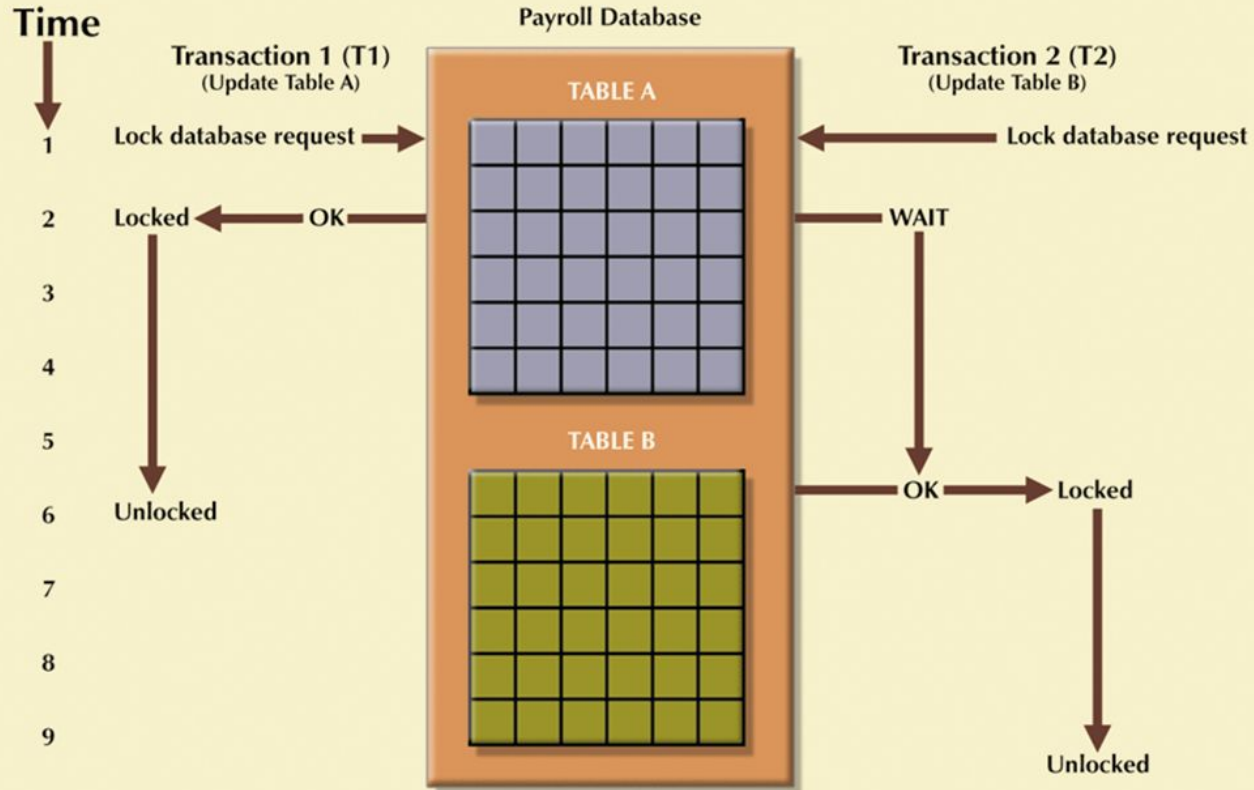


FIGURE 10.5 An example of a page-level lock

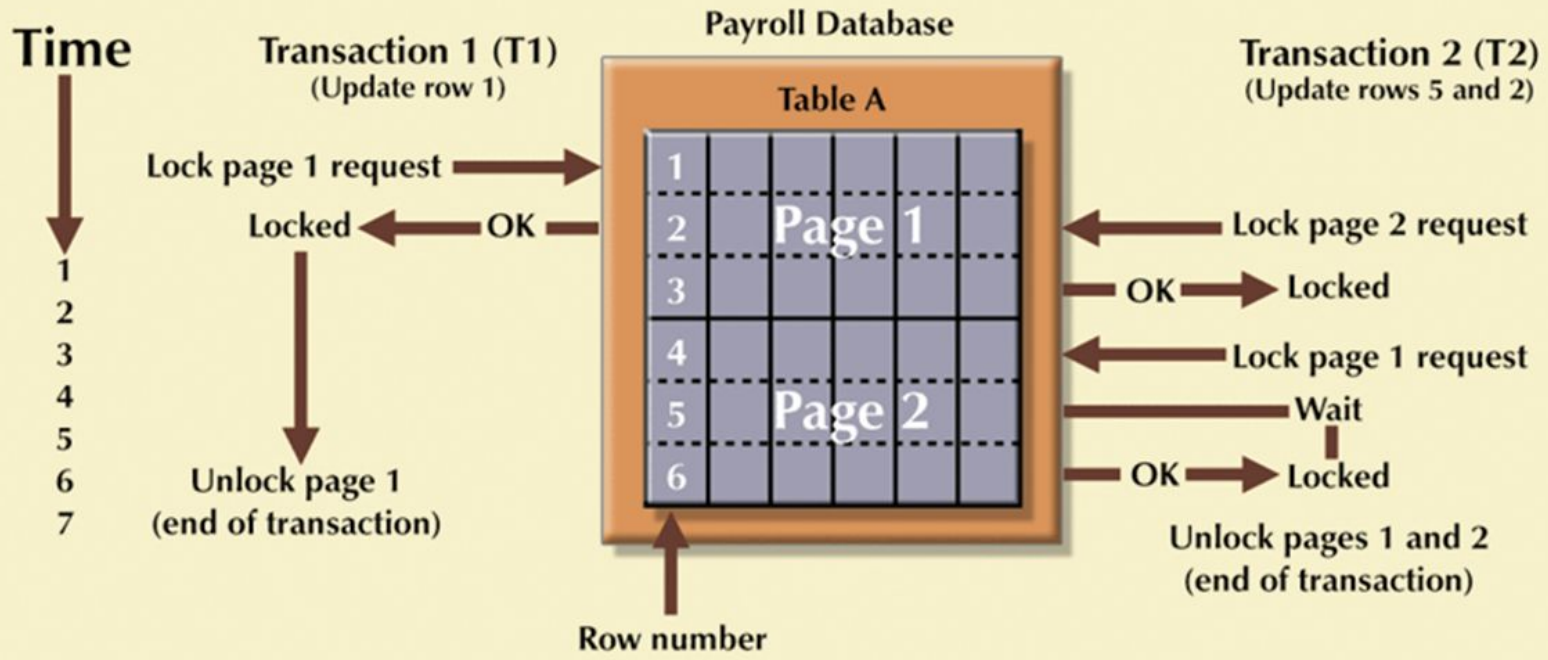
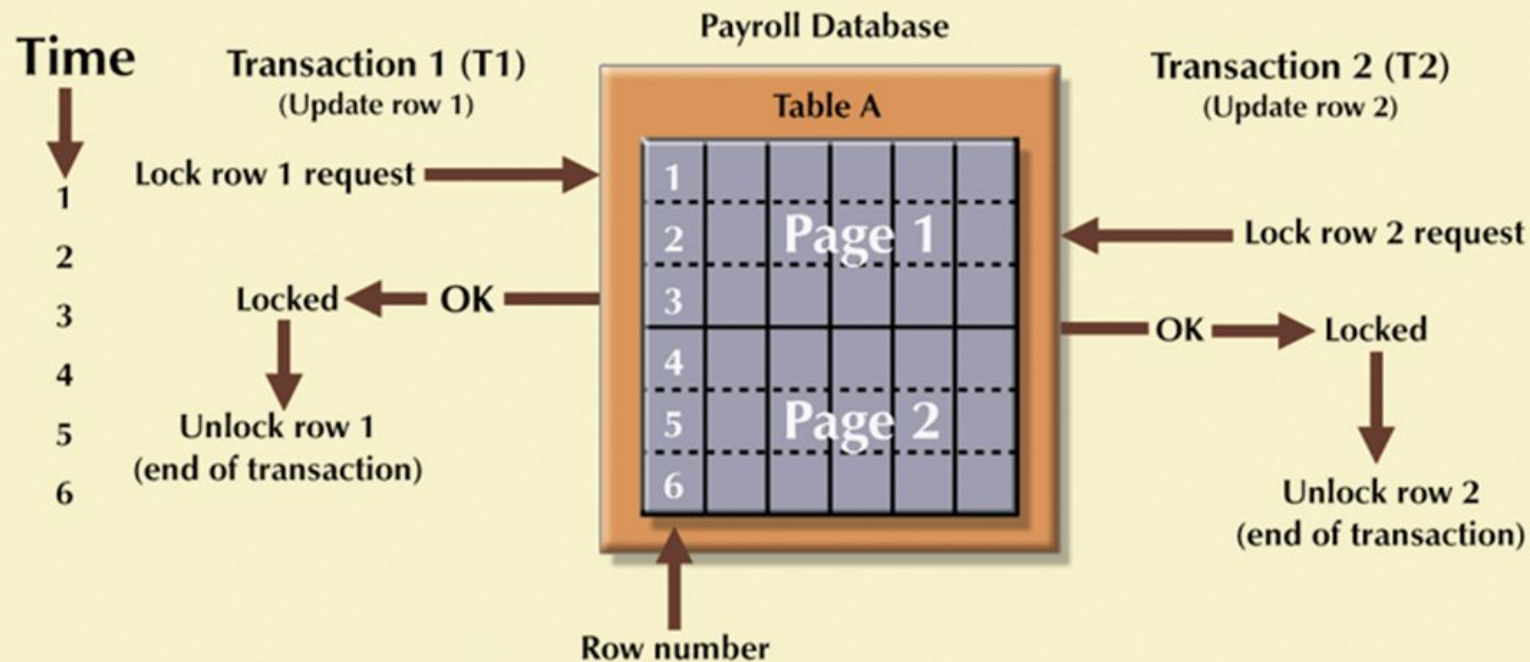


FIGURE 10.6 An example of a row-level lock

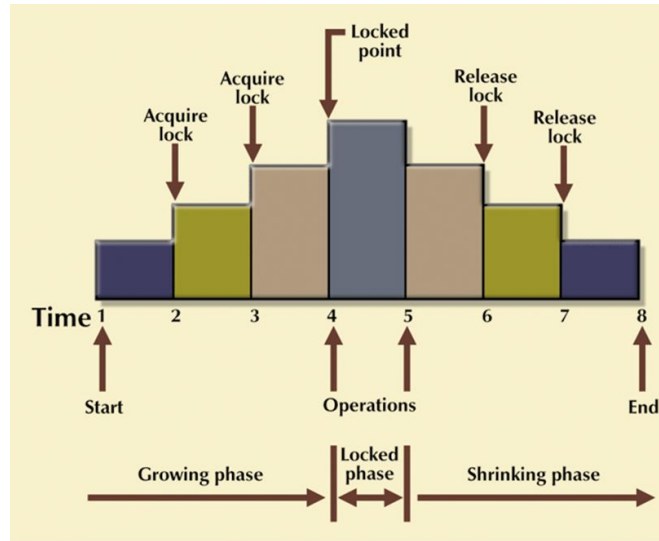


Lock Types

- Binary lock
 - Has only two states: locked (1) or unlocked (0)
- Exclusive lock
 - Access is specifically reserved for transaction that locked object
 - Must be used when potential for conflict exists
- Shared lock
 - Concurrent transactions are granted Read access on basis of a common lock

Two-Phase Locking for Serializability

- Governed by the following rules:
 - Two transactions cannot have conflicting locks
 - No unlock operation can precede a lock operation in the same transaction
 - No data are affected until all locks are obtained—that is, until transaction is in its locked point



Deadlocks

- Condition that occurs when two transactions wait for each other to unlock data
- Possible only if one of the transactions wants to obtain an exclusive lock on a data item
 - No deadlock condition can exist among shared locks

- Can be controlled through:
 - Prevention
 - Detection
 - Avoidance

TIME	TRANSACTION	REPLY	LOCK STATUS	
0			Data X	Data Y
1	T1:LOCK(X)	OK	Unlocked	Unlocked
2	T2: LOCK(Y)	OK	Locked	Unlocked
3	T1:LOCK(Y)	WAIT	Locked	Locked
4	T2:LOCK(X)	WAIT	Locked	Locked
5	T1:LOCK(Y)	WAIT	Locked	Locked
6	T2:LOCK(X)	WAIT	Locked	Locked
7	T1:LOCK(Y)	WAIT	Locked	Locked
8	T2:LOCK(X)	WAIT	Locked	Locked
9	T1:LOCK(Y)	WAIT	Locked	Locked
...				
...				
...				
...				



Concurrency Control with Time Stamping Methods

- Assigns global unique time stamp to each transaction
- Produces explicit order in which transactions are submitted to DBMS
- Uniqueness
 - Ensures that no equal time stamp values can exist
- Monotonicity
 - Ensures that time stamp values always increase
- Additional **wait/die** and **wound/wait schemes**:
 - **Wait/die** : Older transaction waits and younger is rolled back and rescheduled
 - **Wound/wait** : Older transaction rolls back younger transaction and reschedules it

Concurrency Control with Optimistic Methods

- Optimistic approach
 - Based on assumption that majority of database operations do not conflict
 - Does not require locking or time stamping techniques
 - Transaction is executed without restrictions until it is committed
 - Phases are read, validation, and write

Database Recovery Management

- Recover is defined as the following:
 - Restores database from **given state, usually inconsistent, to previously consistent state**
 - Based on **atomic** transaction property
 - All portions of transaction must be treated as **single logical unit of work**, so all operations must be applied and completed to produce consistent database
 - If transaction operation cannot be completed, **transaction must be aborted**, and any changes to **database must be rolled back (undone)**

Transaction Recovery

Makes use of **deferred-write** and **write-through** techniques.

Deferred write

- Transaction operations do not immediately update physical database
- Only transaction log is updated
- Database is physically updated only after transaction reaches its commit point using transaction log information

Write-through

- Database is immediately updated by transaction operations during transaction's execution, even before transaction reaches its commit point

A Transaction Log for Transaction Recovery Examples

*****C*R*A*S*H*****

Thank you for listening!
